

A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 29

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• Demersal trawl: seabed species - cod, haddock, plaice, shrimps, scampi; chains / weights damage the seabed (mix aerobic and anaerobic layers, destroy sea fans, corals, seagrass); high by-catch in mixed fisheries; high fuel use • Purse seine: shoaling pelagic species - tuna, sardines, herring, anchovy; can remove whole shoals; dolphins associated with tuna shoals may be caught • Pelagic long-line: tuna, squid; albatrosses (long-lived, low breeding rate) drown on baited hooks; sharks and turtles caught	6
01.2	• Shrimp trawls use very small mesh so crabs, molluscs, sea urchins, starfish, juvenile fish cannot escape - by-catch can be 99% of the catch • Drift nets are non-selective curtains near the surface catching whales, dolphins, turtles, sharks	2
01.3	• Lost or discarded nets and traps continue to catch and kill organisms; dead animals act as bait attracting more • Biodegradable ropes / panels so lost traps fall apart; radio transmitters on traps so they can be recovered; net recovery schemes	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 340-343; AQA spec 3.5.2.2.	
02.1	• Giant tritons are predators of the crown-of-thorns starfish which eats coral polyps • Removing tritons allowed starfish populations to explode, destroying large areas of reef (food web / trophic cascade)	2

Question	Marking guidance	Marks
02.2	• Fixed mooring buoys so boats do not anchor on coral • Education of divers / snorkellers not to touch coral or kick up sediment; bans on spear fishing and souvenir collection • Fishing controls: no-take zones (Chagos, Great Barrier Reef, Lundy, Lamlash Bay), seasons, minimum and maximum catch sizes, bans on dynamite / small-mesh nets / trawling • Protection of turtle nesting beaches; control of coastal development; protection of mangroves and seagrass; artificial reefs (reef balls, sunk ships / oil rigs)	4
02.3	• No sunlight so no symbiotic algae; polyps rely only on captured food so have far less energy and grow very slowly • Damage from trawls takes hundreds of years to repair; the slow-growing, low-fecundity fish (orange roughy) they support are easily over-exploited	2
02.4	• Deep-water trawling (also by-catch); oil and gas exploration in deeper water; ocean acidification • Protected areas banning trawling, eg Darwin Mounds SAC off NW Scotland, Oculina Bank off Florida	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 61-64; AQA spec 3.1.2.3.4.	
03.1	• Single genes can be transferred from any species, not just closely related ones that produce fertile hybrids • Desired characteristics introduced without linked unwanted ones; faster	2
03.2	• Bt: reduces insecticide spraying so fewer non-target deaths, lower cost / energy • Roundup Ready: weeds controlled by spraying glyphosate without harming the crop, so higher yields with less cultivation / tillage • Golden rice: provides vitamin A in the staple diet, reducing about 500 000 cases of blindness a year in India / sub-Saharan Africa	3
03.3	• Gene transfer by pollen to conventional / organic crops or wild relatives (eg Bt gene killing insects feeding on wild plants) • Antibiotic-resistance marker genes could transfer to pathogenic bacteria; possible allergies; gene transfer to gut bacteria • Detection in the human food chain from crops approved only for animal feed • Economic: seeds must be bought each year from patent-holding companies, raising costs and reducing local crop diversity and seed saving in LEDCs; herbicide-resistant crops may increase herbicide use	4
03.4	• Public / political concern that risks are not fully assessed - more evidence needed (precautionary principle)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 323-324; AQA spec 3.5.1.2.	
04.1	• Hard release: individuals released with no post-release support - suitable for species whose behaviour is instinctive and need no learned skills (insects, fish, amphibians, reptiles) • Soft release: gradual release into larger areas with food provided while they learn to find food, avoid predators - needed for mammals and birds • Problems faced: finding / recognising food and poisonous foods, hunting skills (captive-born predators never learned), avoiding predators, acceptance into wild social groups	3

Question	Marking guidance	Marks
04.2	<i>Any 5 from:</i> • Large enough suitable habitat for a viable population • Reliable food supply • Low predation risk / original threat removed • Suitable breeding sites • Water • Support of the local human population • Official / legal protection of habitat and species	5
04.3	• Tracking (radio / GPS tags, rings) shows survival, dispersal and breeding of released animals so the success of the project can be assessed • Identifies causes of failure so future releases can be improved / extra support given	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 36-37; AQA spec 3.1.2.3.3.	
05.1	• Sieves assembled with the coarsest mesh on top; a dried, crushed soil sample is placed in the top sieve • Stack shaken (about 2 minutes) until no more particles pass; each fraction weighed • Percentages of sand (0.02-2 mm), silt (0.002-0.02 mm) and clay (base pan, below 0.002 mm) calculated; material over 2 mm (stones, twigs) is not soil; plotted on a soil triangle	3
05.2	• Auger collects a core of known cross-sectional area and depth • So samples from different sites are directly comparable (also needed for bulk density = dry mass / volume)	2
05.3	• Minerals such as calcium carbonate break down releasing carbon dioxide • So mass is lost that is not organic matter	2
05.4	• A beam of light passes through the sample in a dark chamber; a light sensor at 90° measures light scattered by suspended particles - more turbid, more scatter • Standardise bulb brightness / colour, distance and container • A calibration curve is produced using samples with known amounts of suspended matter so readings convert to concentrations; Secchi discs / turbidity bottles give only semi-quantitative results	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 410-413, 417; AQA spec 3.7.2.1-3.7.2.2.	
06.1	• Other MEDCs (USA, Canada, Australia) also had surpluses; Japan already had suppliers • Eastern Europe / USSR needed food but could not pay • Selling cheaply in LEDCs would undercut local farmers, put them out of business and reduce their long-term food production	3
06.2	• Quotas limiting production (eg milk) • Set-aside payments for taking land out of production • Farm diversification (recreation, farm shops, specialised products) and alternative crops / livestock (biofuels, pharmaceutical poppies, llamas, deer, bees) • Agri-environmental schemes (ESA, CSS, ESS, CS) paying for environmentally beneficial farming	4

Question	Marking guidance	Marks
06.3	<i>Any 3 from:</i> • Hedgerows at least 1.5 m tall cut at 2-year intervals outside the breeding season - bird / insect habitat and landscape • Buffer strips / field margins (no cultivation, fertiliser or pesticide within 2 m) - protect ditches and rivers, habitat • Beetle banks of tussocky grass - natural pest predators • Skylark nesting plots; wild bird seed / nectar plants - winter food and pollinators • In-field tree protection; controlled stocking density to reduce compaction; ditch management; dry-stone walls; woodland management	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 332-334; AQA spec 3.5.1.4.	
07.1	• SSSI: UK legislation (Wildlife and Countryside Act); best examples of UK flora, fauna, geological features; management plan agreed with owner who must notify Operations Likely to Damage (ploughing, drainage, pesticides, tree planting / removal) • SPA: EU Birds Directive; places of international importance for birds (with SACs under the Habitats Directive form the Natura 2000 network; may restrict activities in surrounding areas) • Ramsar: Ramsar Convention (intergovernmental agreement); internationally important wetlands, originally for waterfowl, now wetland conservation and wise use generally • MCZ: UK marine legislation (Marine and Coastal Access Act) under the W&CA framework; protects marine habitats / species by restricting activities such as fishing and dredging; NNR: best SSSIs with complete communities	4
07.2	• Legal status: restrictions on damaging activities within the area (and sometimes in surrounding areas, eg drainage) • Management agreements between owners and statutory bodies to maintain suitable conditions • But it does not protect against external influences such as air pollution or climate change; very few sites are state-owned so protection depends on owner cooperation	3
07.3	• Much of the UK's wildlife lives in habitats created by farming (hay meadows, hedgerows) which only survive if the farming continues • Managing land for wildlife costs money / reduces income so farmers need payments (points for beetle banks, hedgerows, buffer strips, wild bird seed, low-input grassland) • Benefits everyone so society pays through central funding; legislation cannot compel positive management	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 30, 38-39; AQA spec 3.1.2.3.2.	
08.1	• Moral belief: protecting the environment is the 'right' thing to do regardless of benefit to people • Enlightened self-interest: protecting the environment is done because it causes fewer difficulties (for people) than not doing so	2
08.2	• Total impact = population size x the individual (per capita) impact of each person • Per capita impact rises with affluence / access to energy, so both a larger population and a more affluent population increase the total impact	2

Question	Marking guidance	Marks
08.3	<i>Any 1 from:</i> • Slash and burn agriculture • Fishing • Discharge of sewage / biodegradable pollutants	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370; AQA spec 3.6.1.	
09.1	• Energy: low energy-density renewable (solar) / non-renewable fossil fuels • Materials: cyclic - wastes of one process are raw materials of the next / linear - used once then discarded, depletion • Wastes: biodegradable, non-toxic or do not build up / toxic, persistent (eg PCBs, organochlorines, toxic metals) that bioaccumulate • Temperature: low / high (Haber process, incineration)	4
09.2	• Abundant simple raw materials build into monomers and then a wide range of polymers (carbohydrates, proteins) • They are re-used in cycles so there is no depletion of raw materials or build-up of wastes	2
09.3	• Use often disperses the minerals (eg in products spread over the world / corrosion) • Or produces mixtures of materials that are difficult to separate, so sustainable exploitation is reduced	2
09.4	• They cannot easily be removed from end-of-life appliances and are not biodegradable • Natural wastes are non-toxic / broken down to raw materials for other processes, whereas persistent toxins may bioaccumulate and biomagnify	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373-374; AQA spec 3.6.2-3.6.3.	

Question	Marking guidance	Marks
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Succession: pioneers, seres (hydrosere, lithosere, psammosere), modification of abiotic conditions, climax communities controlled by climate; deflected succession and plagioclimaxes • Application: deciding whether to manage (plagioclimax habitats need grazing, mowing, burning, coppicing, pollarding continued) or leave alone (climax habitats); habitat creation planning the stages; controlling scrub / secondary succession; understanding that new habitats change over time • Population dynamics: birth and death rates, density-dependent and independent factors, carrying capacity, sigmoid growth, predator-prey cycles, r- and k-selection, MSY • Application: setting sustainable harvest levels (fisheries, whaling, forestry); recognising k-selected species need strict protection and recover slowly; deciding when culling is needed (deer without wolves, introduced species) or when releases / captive breeding are needed (low breeding rate); predicting effects of reintroducing predators • Monitoring population size, distribution, survival rate and age structure to assess success; adaptation and range of tolerance guiding abiotic management • Evaluation: uncertainties (natural fluctuations, unknown carrying capacity); examples (Yellowstone wolves, Scottish deer, chalk grassland); judgement	9
10.2	• Any taxonomic group of organisms classified together by similarity (eg species, genus, family)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 72-85; AQA spec 3.1.3.	

Question	Marking guidance	Marks
11.1	Discuss the extent to which the principles of the circular economy could be used to develop sustainable lifestyles. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Definition of sustainability (Brundtland); population x per capita impact; dynamic equilibria, feedback, tipping points, diversity • Principles of natural systems: renewable low-density energy, low temperatures, material cycles, non-toxic wastes; contrast with linear human systems • Circular economy principles: cycling materials, renewable energy, supporting ecosystems, separation of technical and biological materials, diversity, connected systems, design for extended use and re-use, optimum production, new technologies • Application to land use, water, minerals, waste, pollution, energy, food with examples (Kalundborg, Eigg, Ol Pejeta, Rio Bravo, EU initiatives, DC grids) • Measuring progress: ecological footprint, biocapacity, global hectares, Earth Overshoot Day, carbon footprint, Living Planet Index • Barriers: economics, competition, consumer behaviour, need for international agreement, existing infrastructure • Judgement on the extent achievable Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-389; AQA spec 3.6.	25

Question	Marking guidance	Marks
11.2	Agriculture involves the control of abiotic and biotic factors to maximise food production. Discuss how this control can be achieved and evaluate its environmental impacts. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Species selection: market, range of tolerance, controllability • Abiotic control: temperature, light, water, soil fertility / nutrients, aeration, salinity, pH, CO₂, topography, relief, wind - methods and examples • Biotic control: pest groups, cultural control, biological control (successes and failures), sterile males, pheromones, genetic resistance, pesticides (properties), antibiotics, pollinators, soil biota • Manipulation of food species: population density, monocultures, selective breeding, asexual reproduction, cross-breeding, AI / embryo transfer, GM; energetics - intensive vs extensive, energy subsidies, ratios, FCR, food chain losses • Impacts: habitat loss / change, reduced biodiversity, introduced species, pesticide and nutrient pollution, GHGs, hydrological changes, soil erosion; positive: plagioclimax habitats • Social, economic and political influences shaping methods • Sustainable strategies and evaluation; conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 306-337; AQA spec 3.5.1.	25